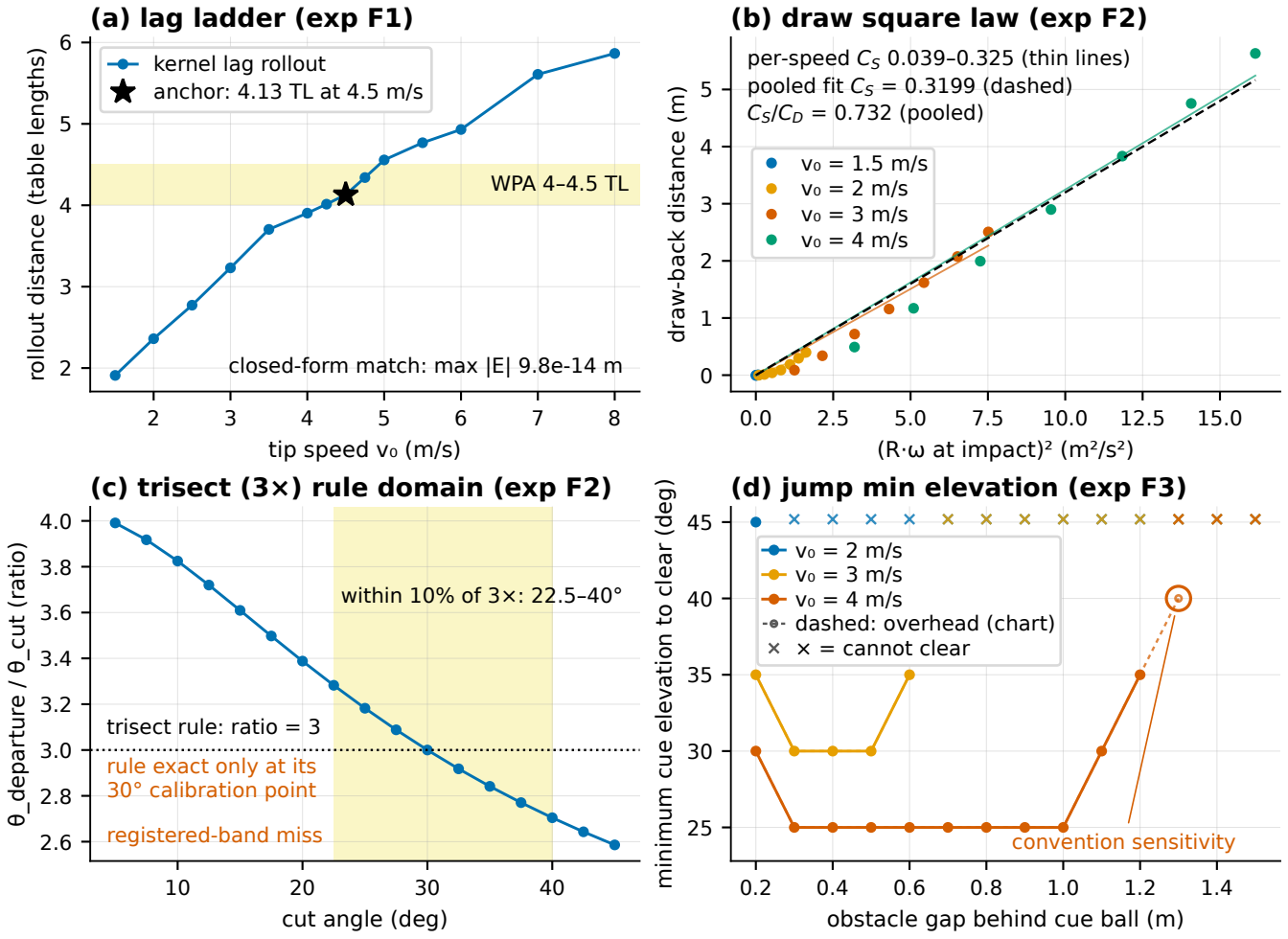
(a) lag ladder (exp F1)
kernel lag rollout
anchor: 4.13 TL at 4.5 m/s
WPA 4–4.5 TL
closed-form match: max |E| 9.8e-14 m
rollout distance (table lengths)
tip speed v₀ (m/s)
(b) draw square law (exp F2)
per-speed C_S 0.039–0.325 (thin lines)
pooled fit C_S = 0.3199 (dashed)
C_S/C_D = 0.732 (pooled)
v₀ = 1.5 m/s
v₀ = 2 m/s
v₀ = 3 m/s
v₀ = 4 m/s
draw-back distance (m)
(R·ω at impact)² (m²/s²)
(c) trisect (3×) rule domain (exp F2)
within 10% of 3×: 22.5–40°
trisect rule: ratio = 3
rule exact only at its
30° calibration point
registered-band miss
θ_departure / θ_cut (ratio)
cut angle (deg)
(d) jump min elevation (exp F3)
v₀ = 2 m/s
v₀ = 3 m/s
v₀ = 4 m/s
dashed: overhead (chart)
× = cannot clear
convention sensitivity
minimum cue elevation to clear (deg)
obstacle gap behind cue ball (m)